

- 1 The stinging nettle, *Urtica dioica*, is a plant species that is commonly found in many areas of the world. Stinging nettles grow in woodland, in gardens and in farmland.

Figure 1.1 shows a stinging nettle.



Figure 1.1

A student decided to investigate photosynthesis in stinging nettles.

- (a) The student made a leaf extract containing chloroplast pigments by grinding some leaves from stinging nettles. The student used chromatography to identify the chloroplast pigments present in the leaves.

Figure 1.2 shows the chromatogram.

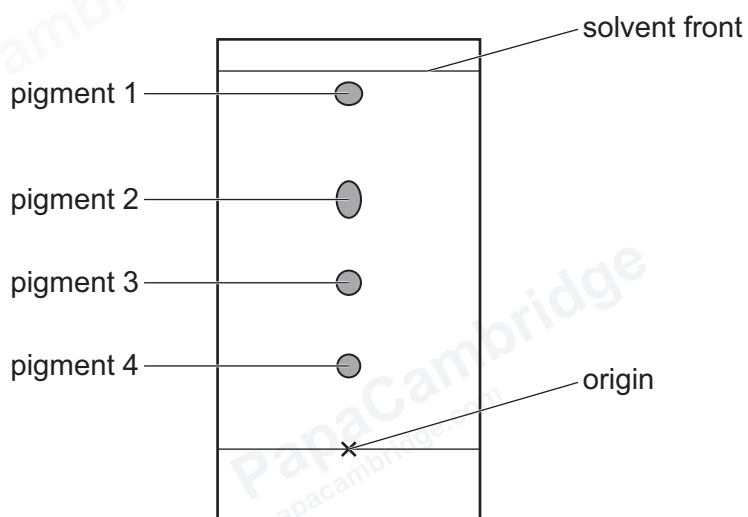


Figure 1.2

Table 1.1 shows the reference R_f values of some chloroplast pigments from a published source.

Table 1.1

chloroplast pigment	R_f value
carotene	0.95
chlorophyll <i>a</i>	0.65
chlorophyll <i>b</i>	0.45
phaeophytin	0.83
xanthophyll	0.71

Use Figure 1.2 and Table 1.1 to identify pigment 1 **and** pigment 2.

pigment 1

pigment 2

[2]

(b) (i) The student read that different light intensities result in different combinations of chloroplast pigments present in the leaves of stinging nettles.

- direct sunlight, with high light intensity
- the shade, with low light intensity.

- collect leaves from stinging nettles growing in direct sunlight **and** in the shade
- use chromatography to separate the chloroplast pigments present in the leaves
- collect the data needed to identify each chloroplast pigment.

You should **not** include a risk assessment.

This image shows a full page of handwriting practice paper. It features multiple sets of horizontal dotted lines spaced evenly down the page, providing a guide for letter height and placement. The background is white, and there are no other markings or text present.



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- (ii) Trichomes are small hair-like structures on the stems and leaves of the stinging nettle that release irritants when they are touched.

The student decided to wear gloves to protect their skin from irritation when touching stinging nettles.

State **one other** hazard, the risk associated with that hazard **and** the safety precaution that the student should take when carrying out their investigation.

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[Total: 11]